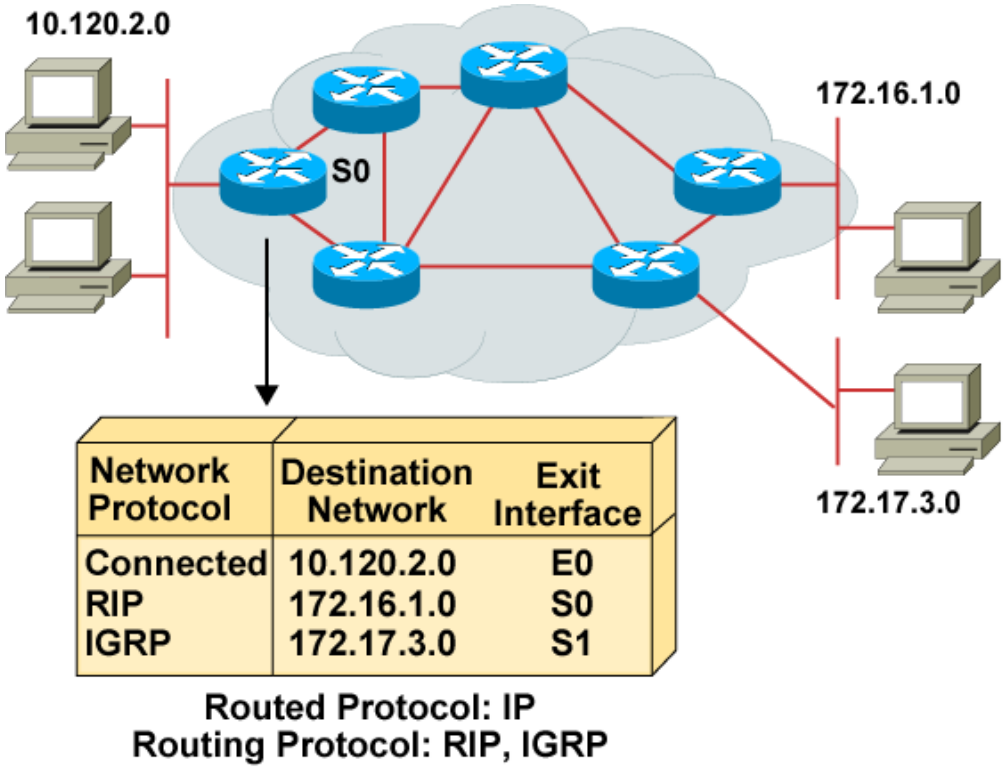
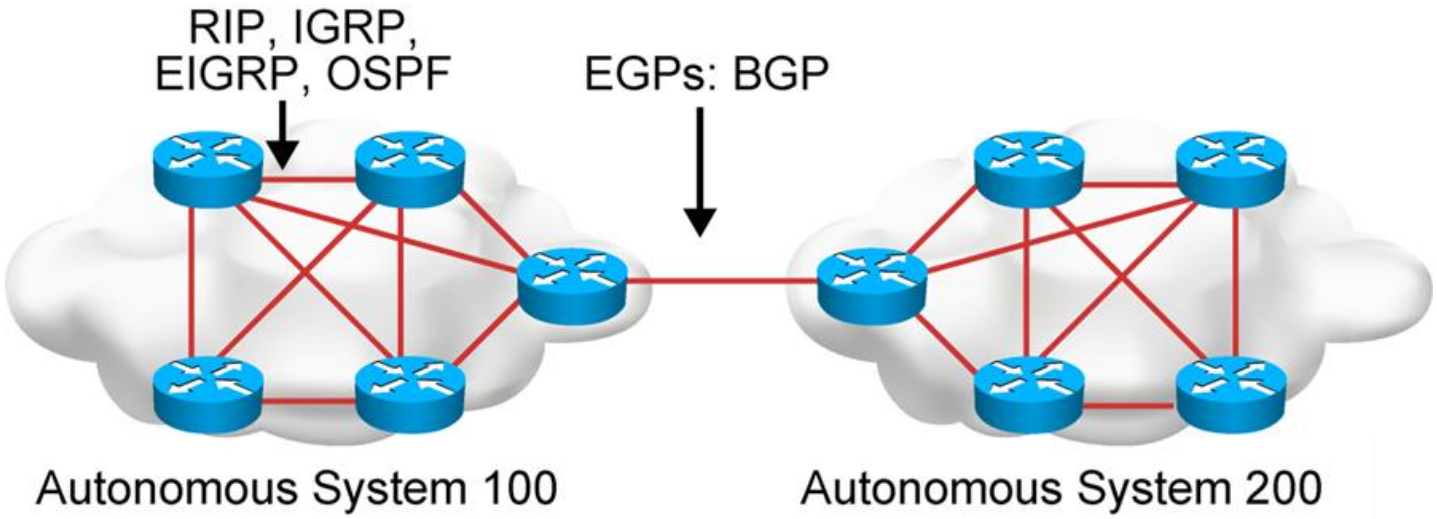
Routing Information Protocol (RIP)

What Is a Routing Protocol ?

- Routing protocols are used between routers to determine paths and maintain routing tables.
- After the path is determined, a router can route a routed protocol.



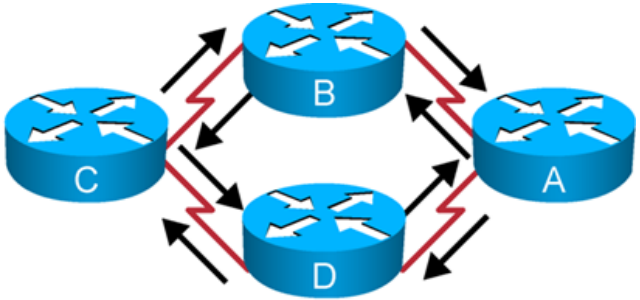
Autonomous Systems: Interior or Exterior Routing Protocols



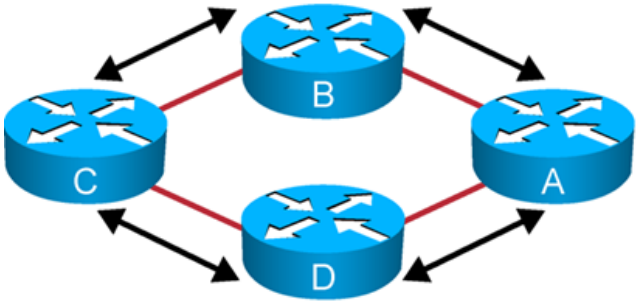
- An autonomous system is a collection of networks under a common administrative domain.
- IGPs operate within an autonomous system.
- EGPs connect different autonomous systems.

Classes of Routing Protocols

Distance Vector

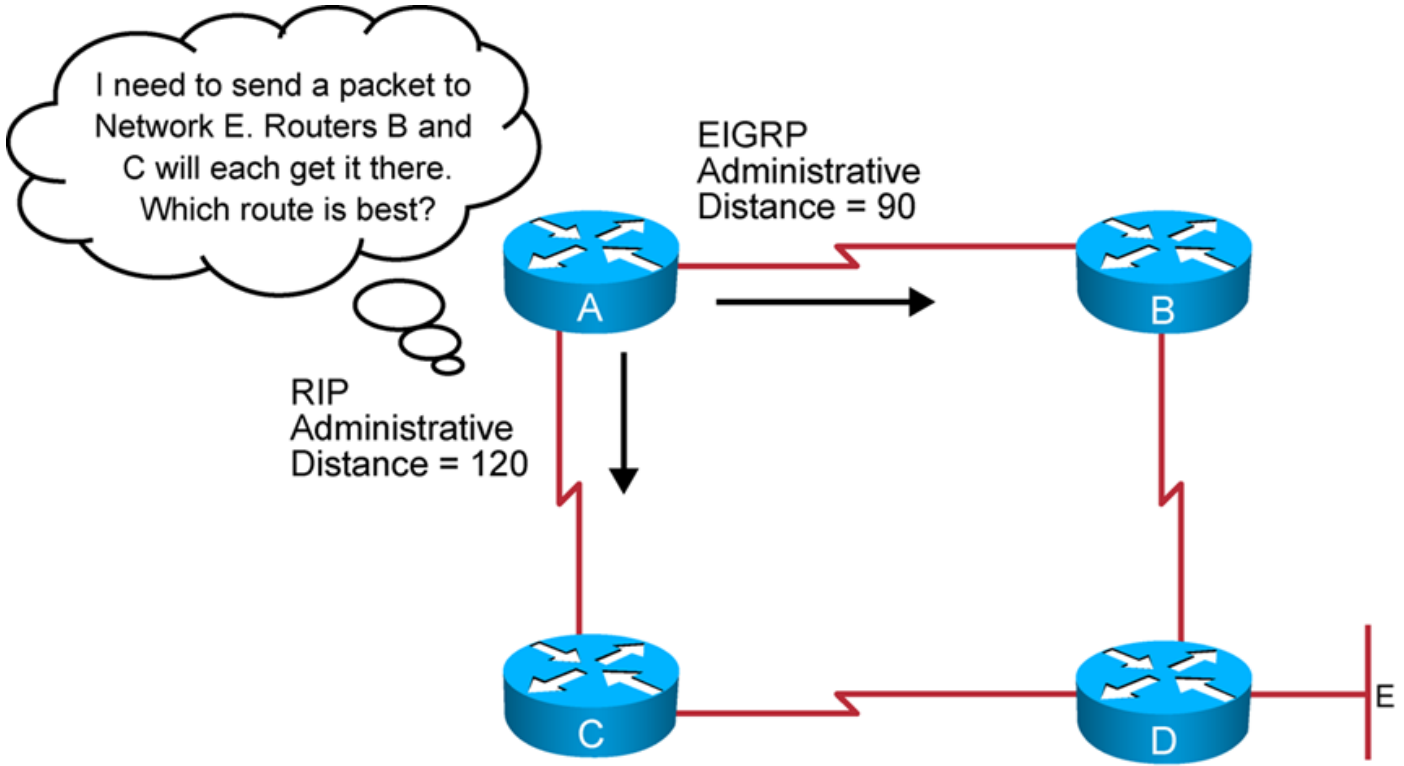


Hybrid Routing

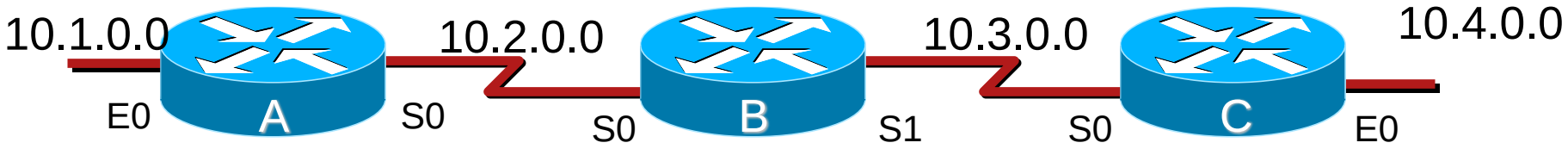


Link-State

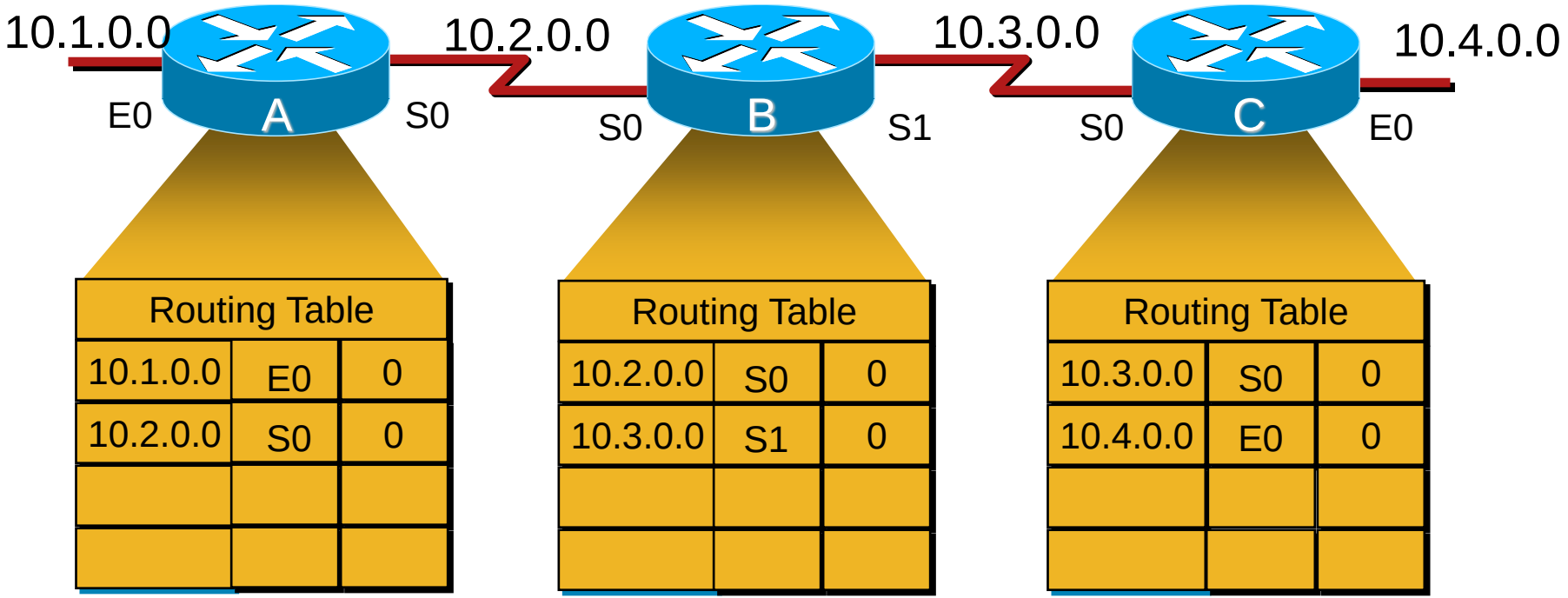
Administrative Distance: Ranking Routes



Distance Vector — Sources of Information and Discovering Routes

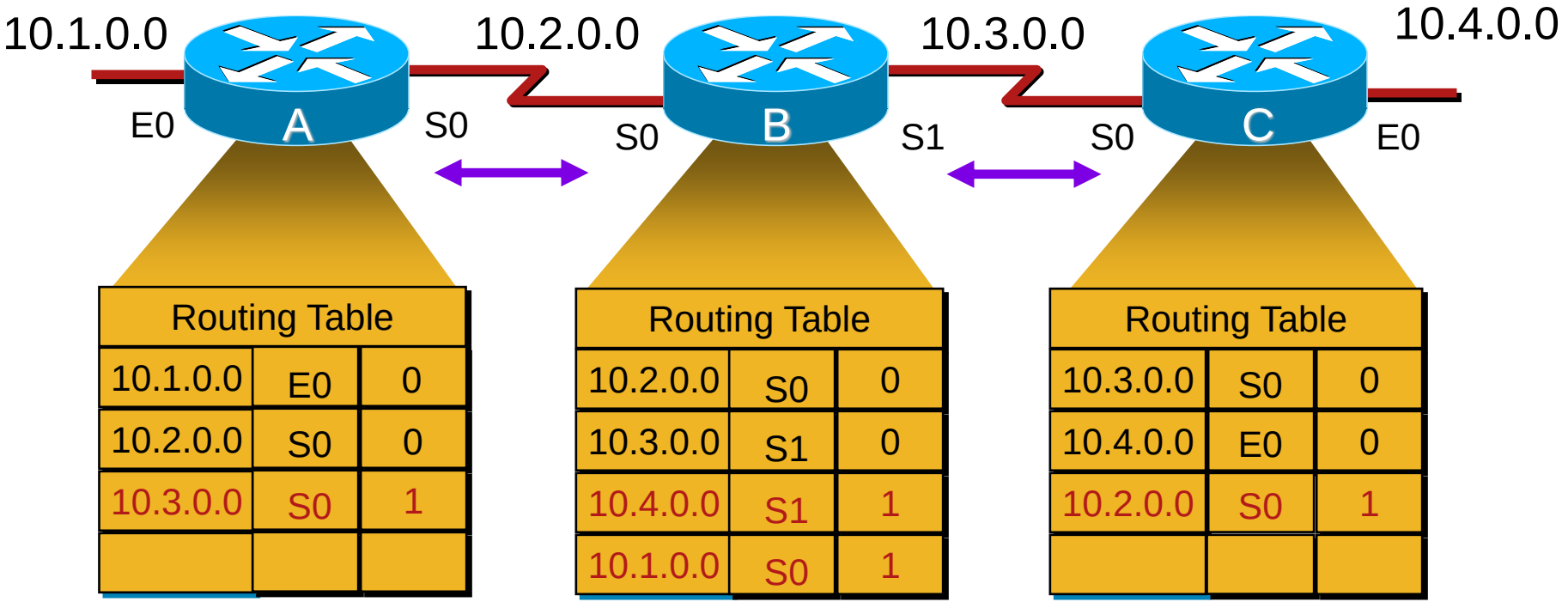


Distance Vector — Sources of Information and Discovering Routes



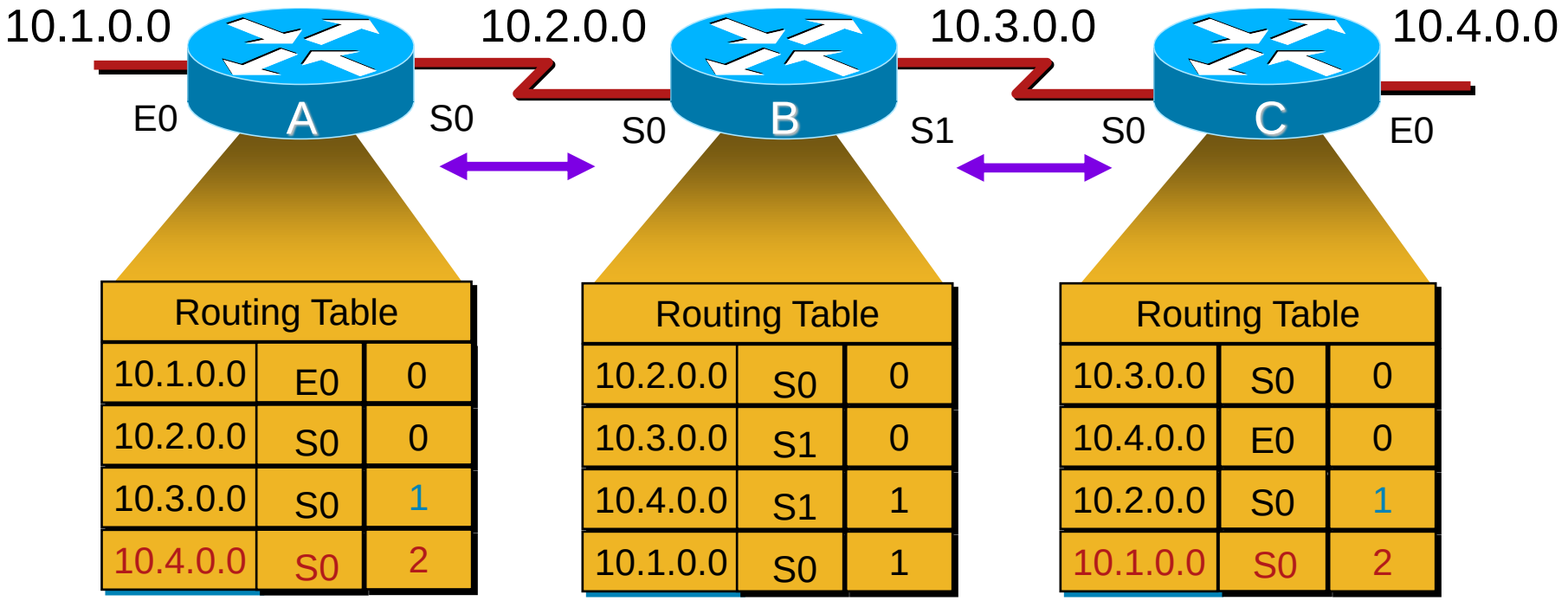
- Routers discover the best path to destinations from each neighbor.

Distance Vector — Sources of Information and Discovering Routes



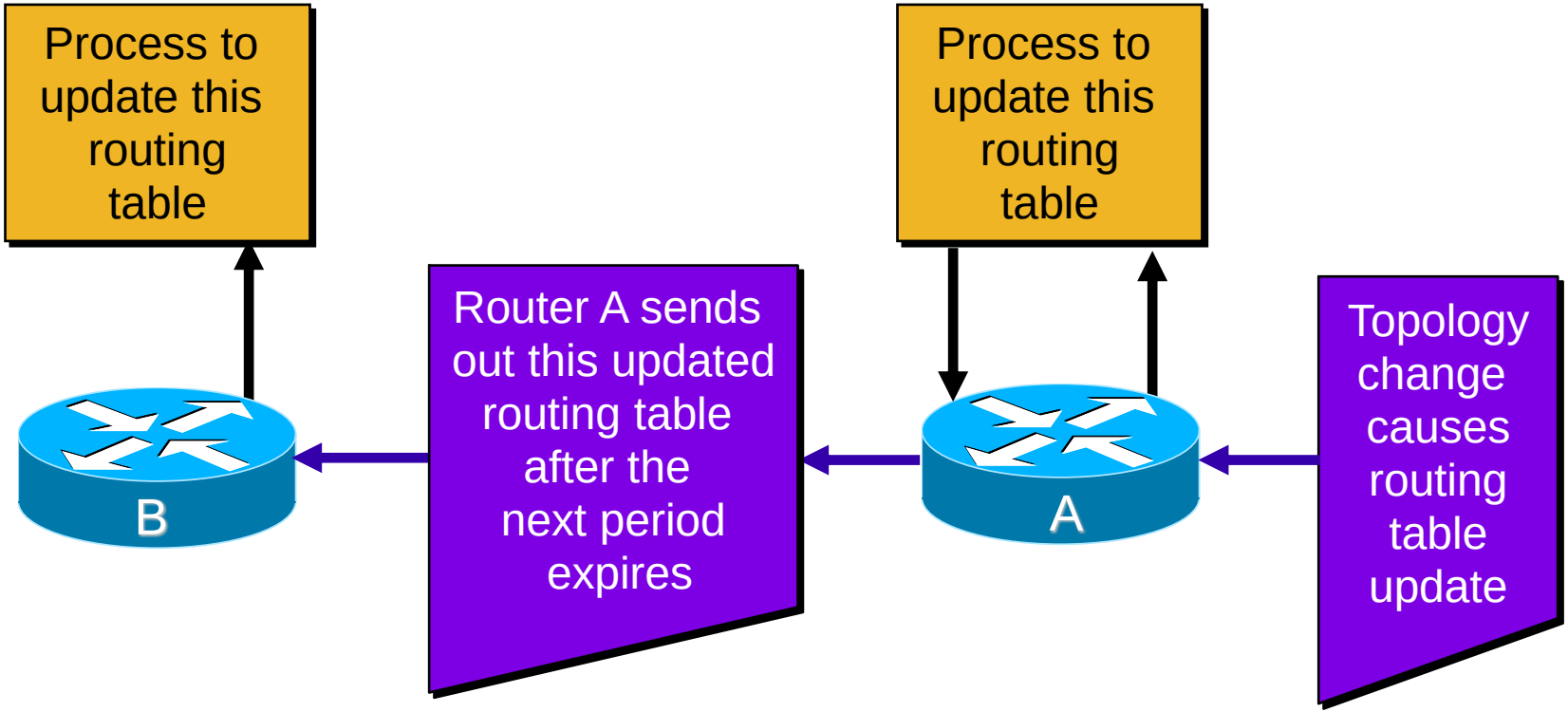
- Routers discover the best path to destinations from each neighbor

Distance Vector — Sources of Information and Discovering Routes



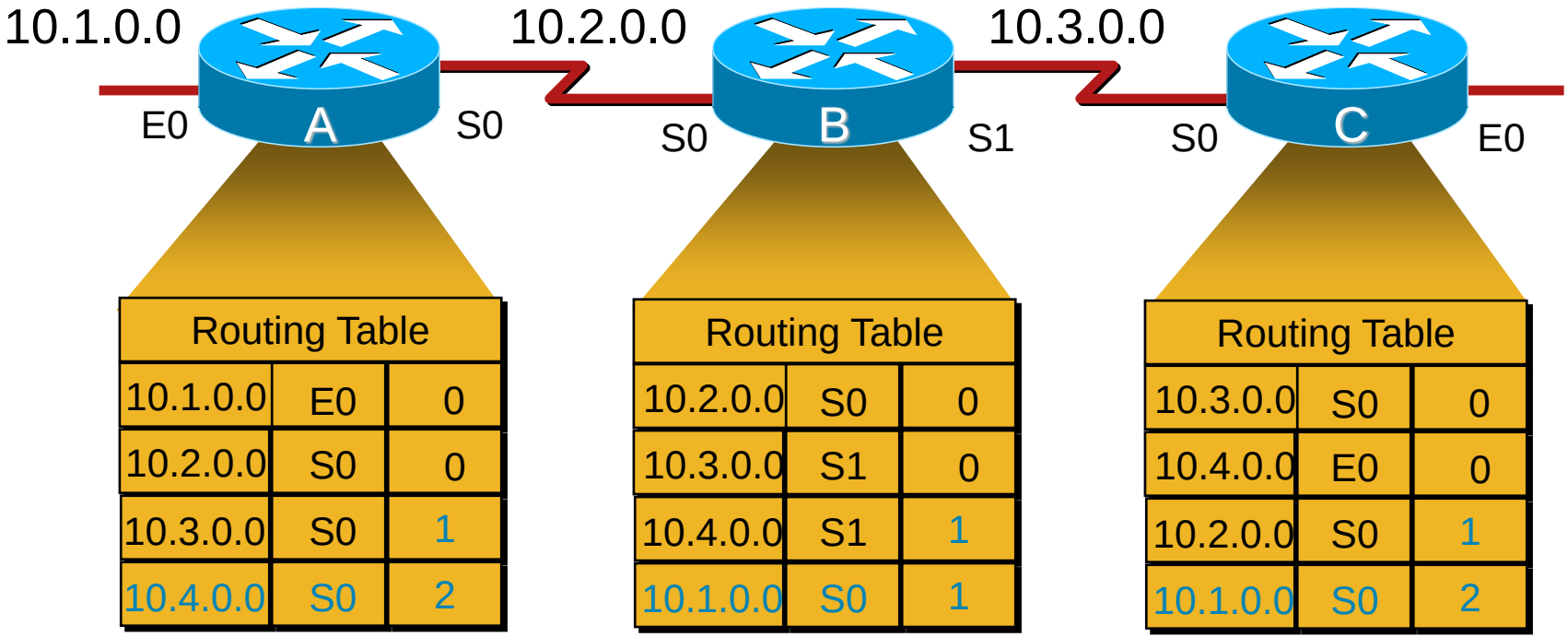
- Routers discover the best path to destinations from each neighbor

Distance Vector — Maintaining Routing Information



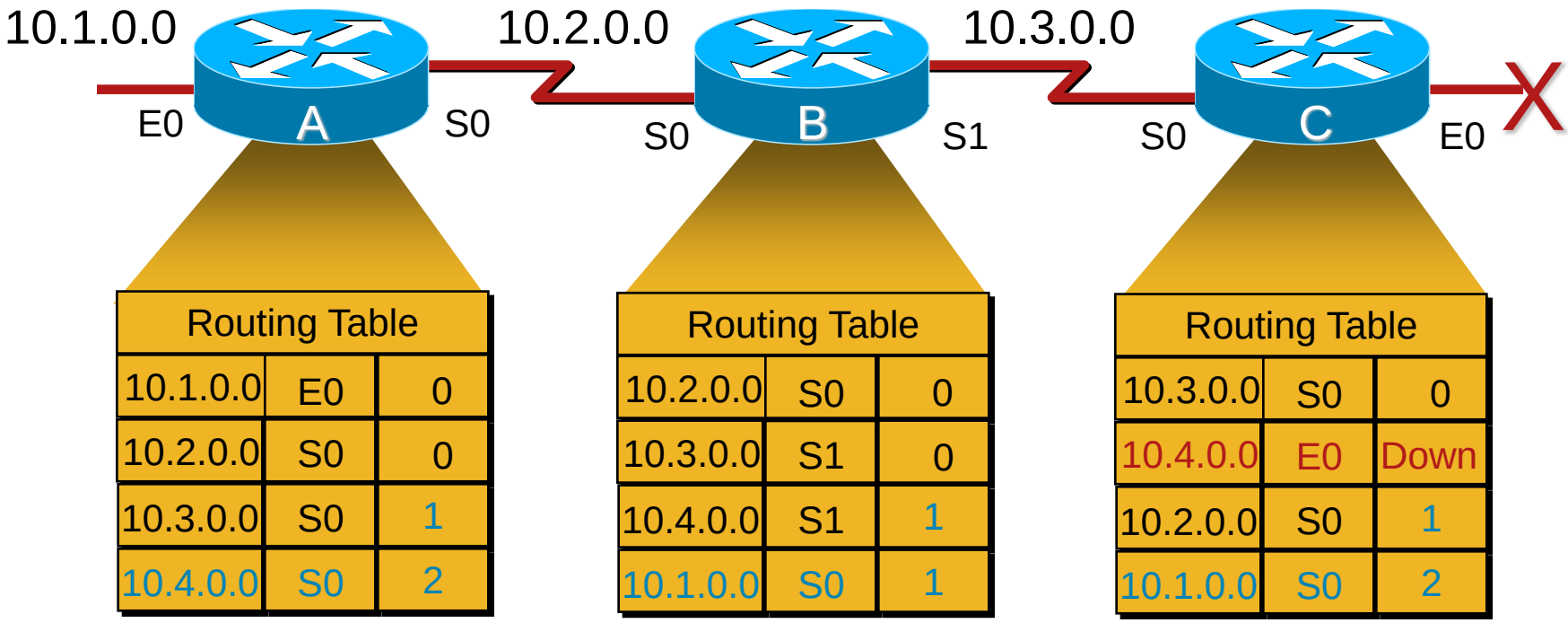
- Updates proceed step-by-step from router to router

Maintaining Routing Information Problem — Routing Loops



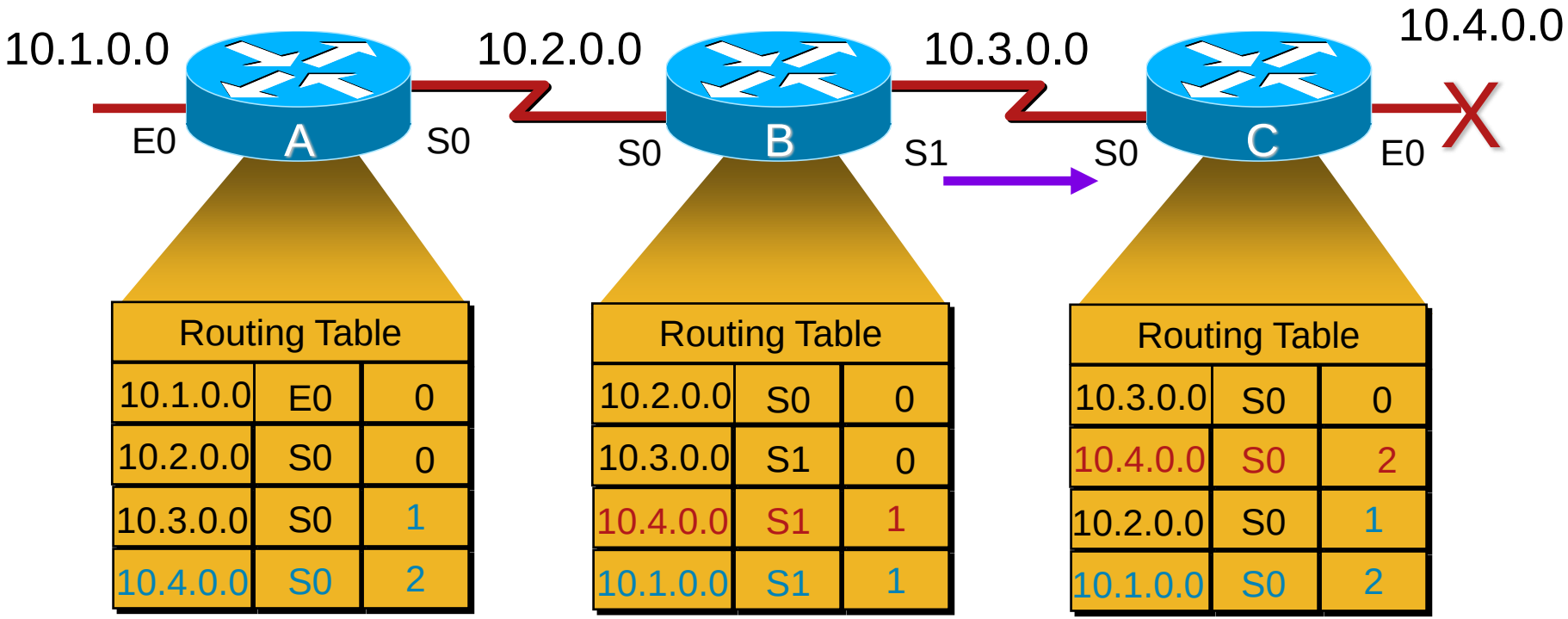
- Each node maintains the distance from itself to each possible destination network.

Maintaining Routing Information Problem — Routing Loops (Cont.)



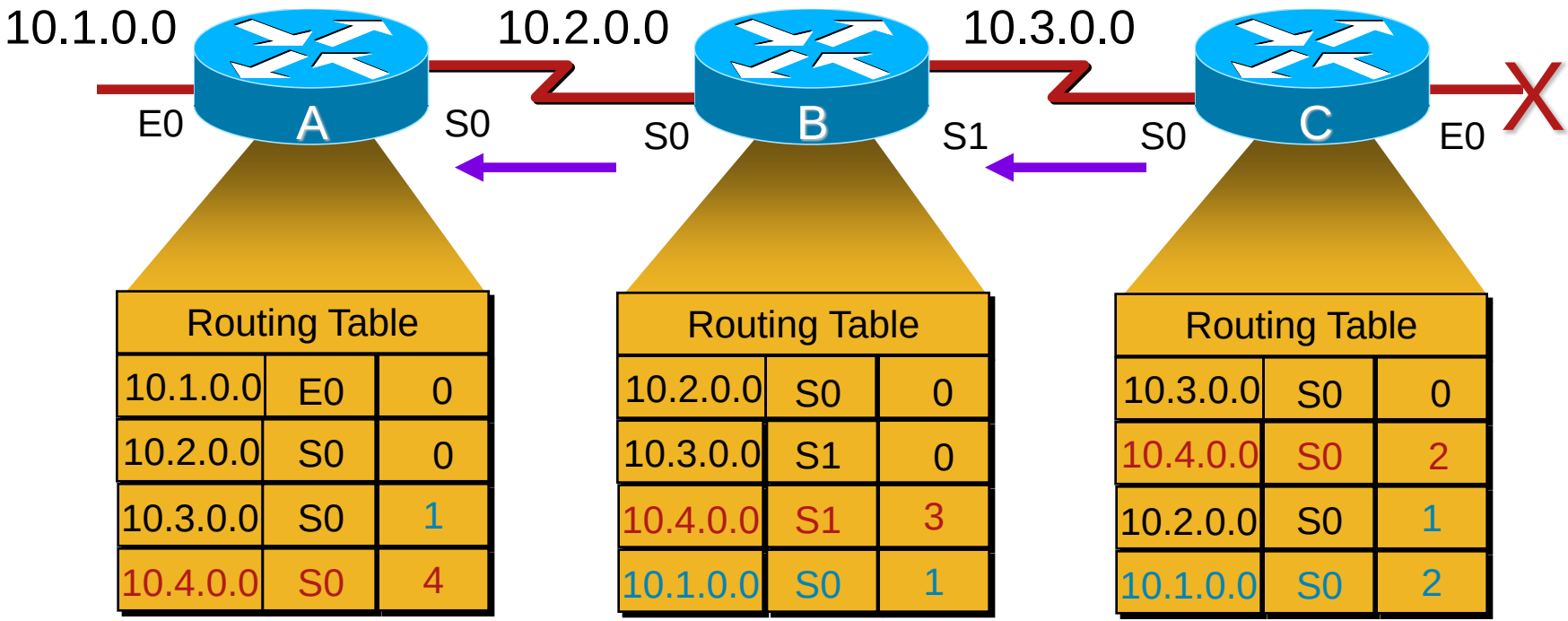
- Slow convergence produces inconsistent routing

Maintaining Routing Information Problem — Routing Loops (Cont.)



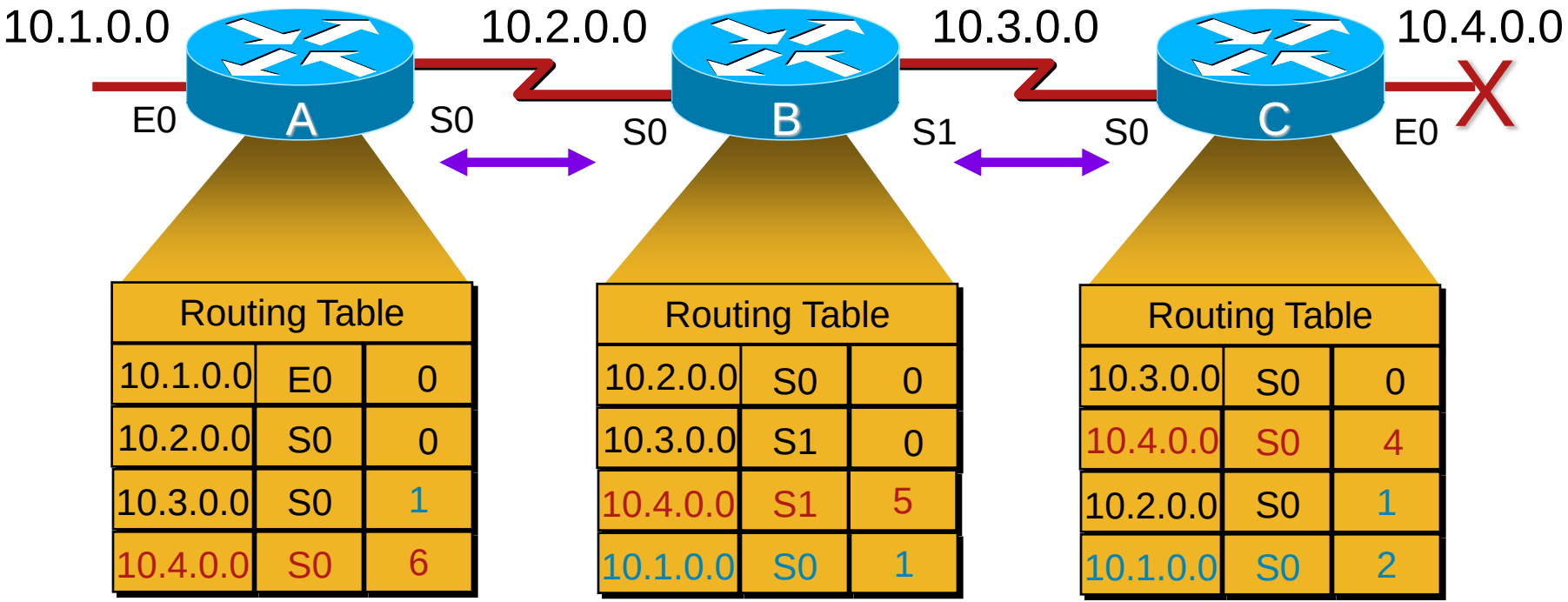
- Router C concludes that the best path to network 10.4.0.0 is through Router B

Maintaining Routing Information Problem — Routing Loops (Cont.)



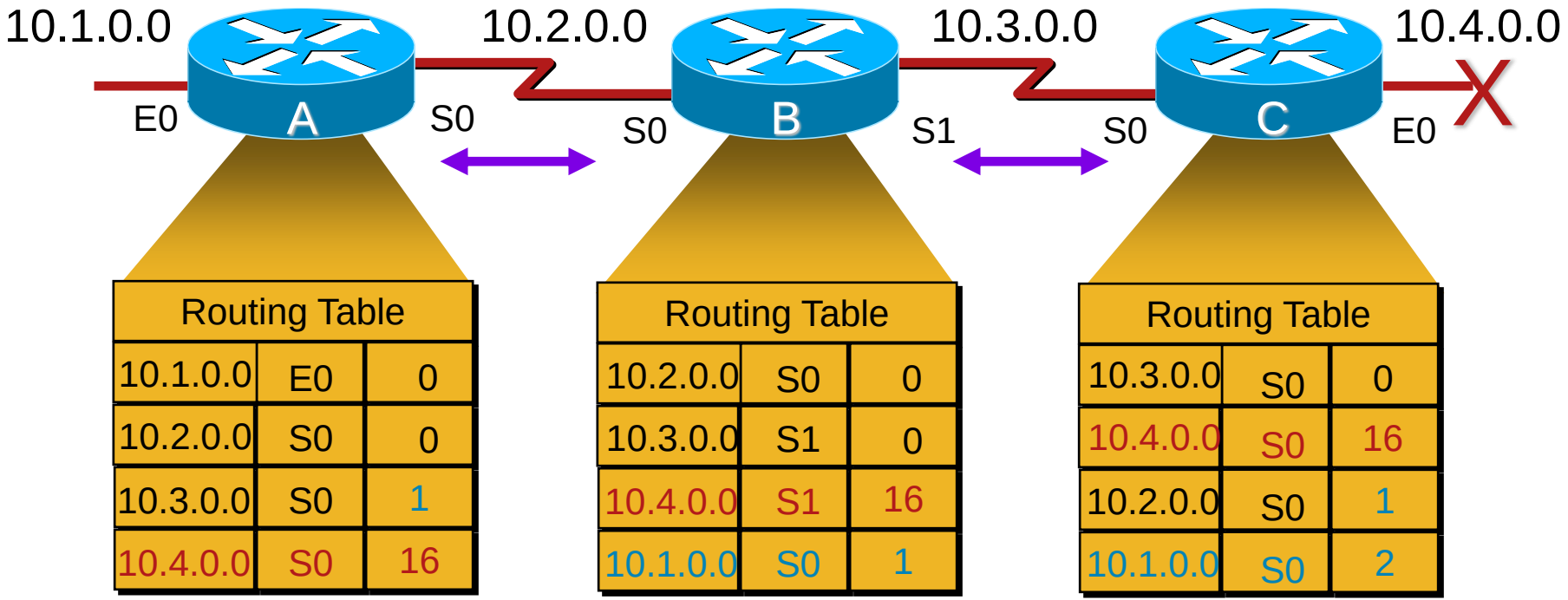
- Router A updates its table to reflect the new but erroneous hop count.

Symptom: Counting to Infinity



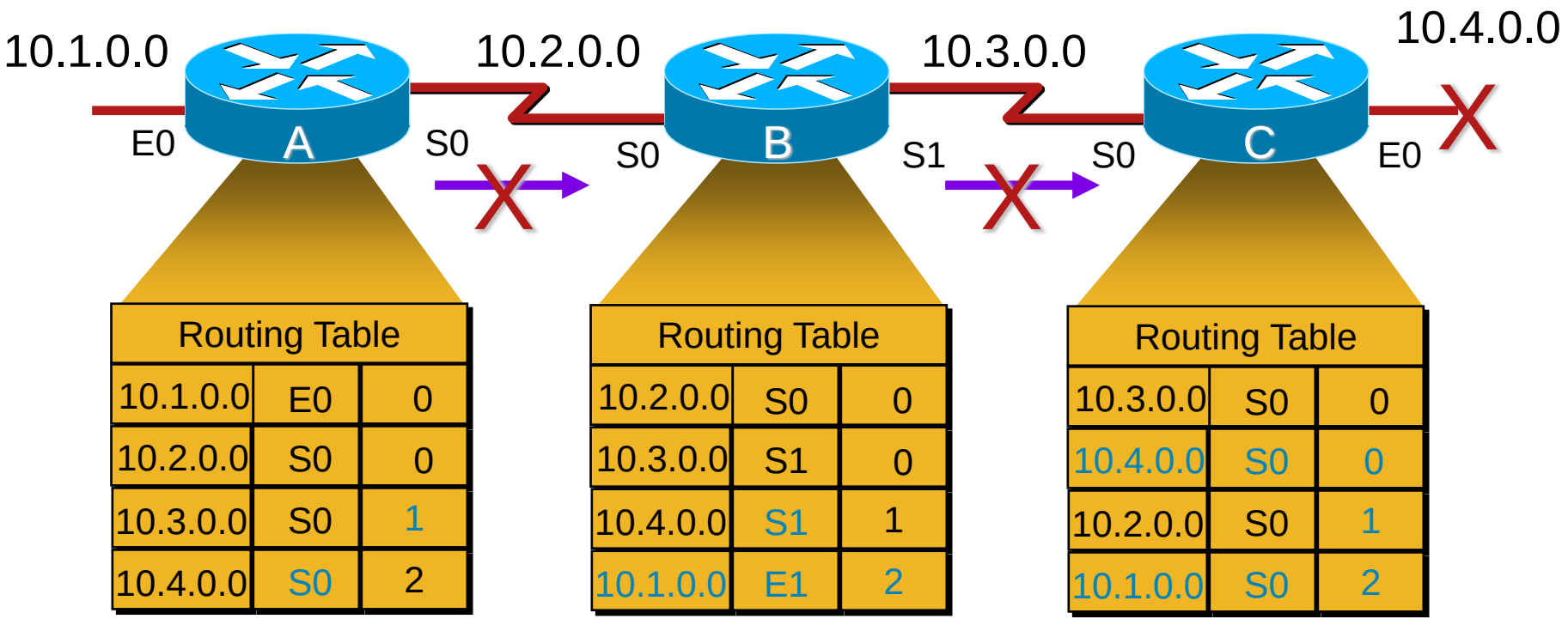
- Packets for network 10.4.0.0 bounce between routers A, B, and C.
- Hop count for network 10.4.0.0 counts to infinity.

Solution: Defining a Maximum



- Define a limit on the number of hops to prevent infinite loops

Solution: Split Horizon



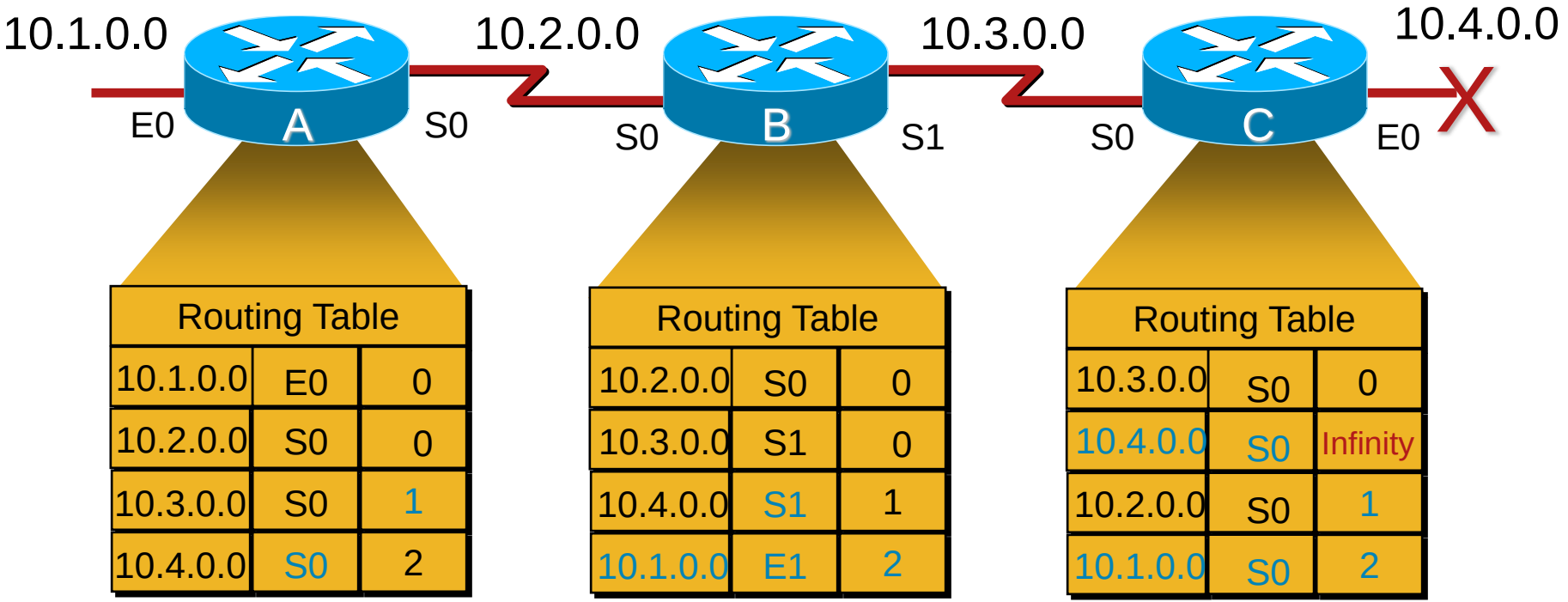
Routing Table		
10.1.0.0	E0	0
10.2.0.0	S0	0
10.3.0.0	S0	1
10.4.0.0	S0	2

Routing Table		
10.2.0.0	S0	0
10.3.0.0	S1	0
10.4.0.0	S1	1
10.1.0.0	E1	2

Routing Table		
10.3.0.0	S0	0
10.4.0.0	S0	0
10.2.0.0	S0	1
10.1.0.0	S0	2

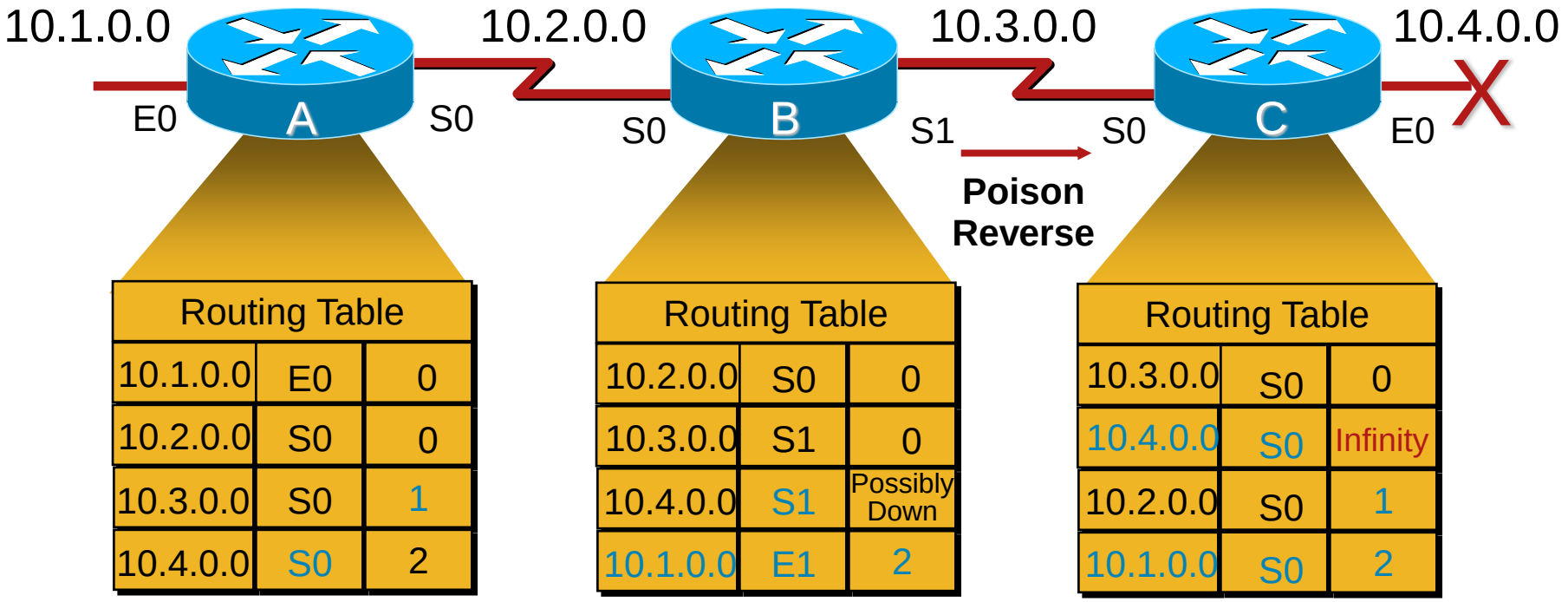
- It is never useful to send information about a route back in the direction from which the original packet came.

Solution: Route Poisoning



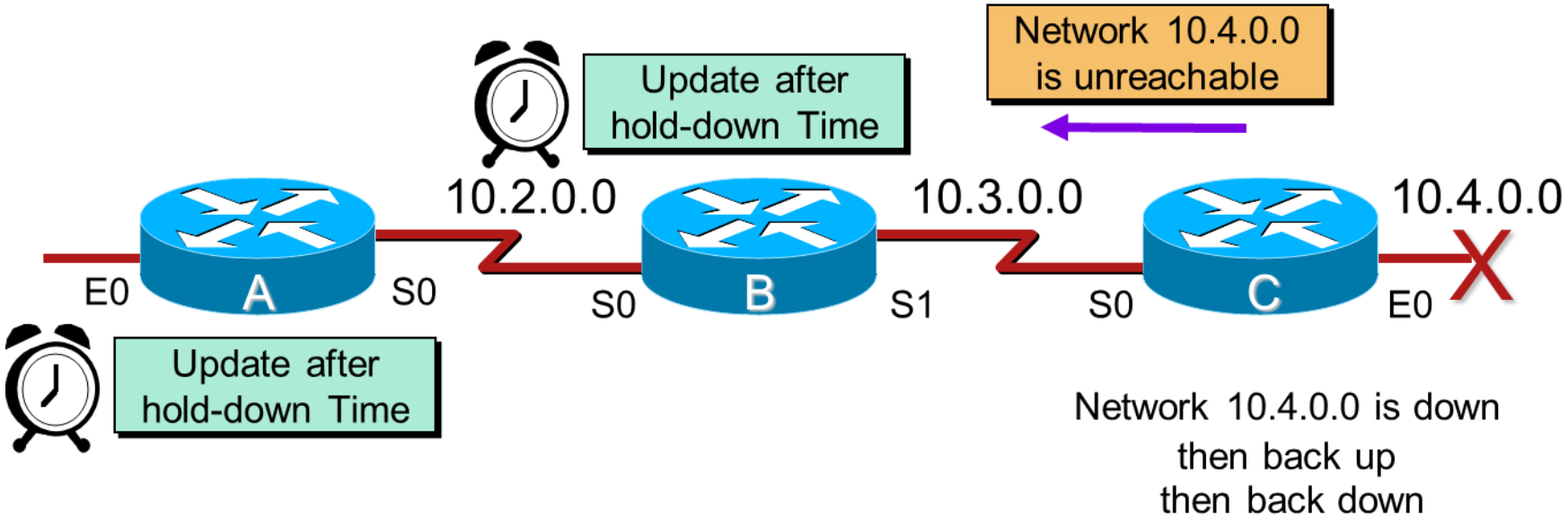
- Routers set the distance of routes that have gone down to infinity.

Solution: Poison Reverse



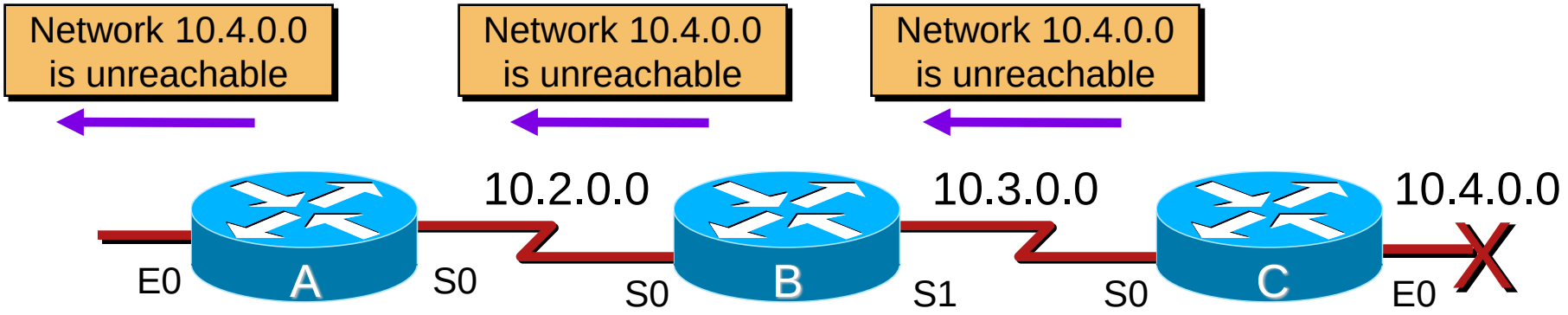
- Poison Reverse overrides split horizon.

Solution: Hold-Down Timers



- Router keeps an entry for the network possibly down state, allowing time for other routers to recompute for this topology change.

Solution: Triggered Updates



- Router sends updates when a change in its routing table occurs.

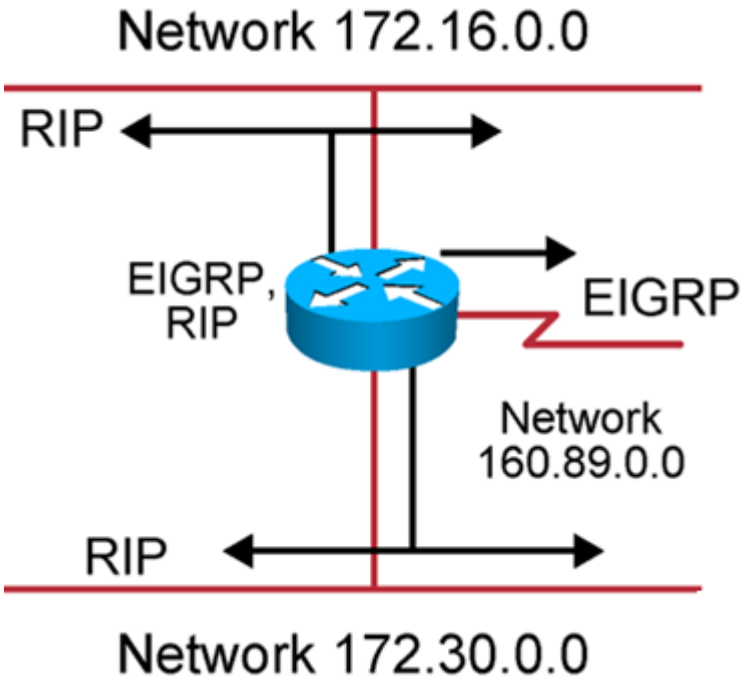
RIPv1 and RIPv2 Comparison

	RIPv1	RIPv2
Routing protocol	Classful	Classless
Supports variable-length subnet mask ?	No	Yes
Sends the subnet mask along with the routing update?	No	Yes
Addressing type	Broadcast	Multicast
Defined in …	RFC 1058	RFCs 1721, 1722, and 2453
Supports manual route summarization ?	No	Yes
Authentication support?	No	Yes



IP Routing Configuration Tasks

- Router configuration
  - Select routing protocols
  - Specify networks or interfaces



RIP Configuration

```
RouterX(config)# router rip
```

- Starts the RIP routing process

```
RouterX(config-router)# version 2
```

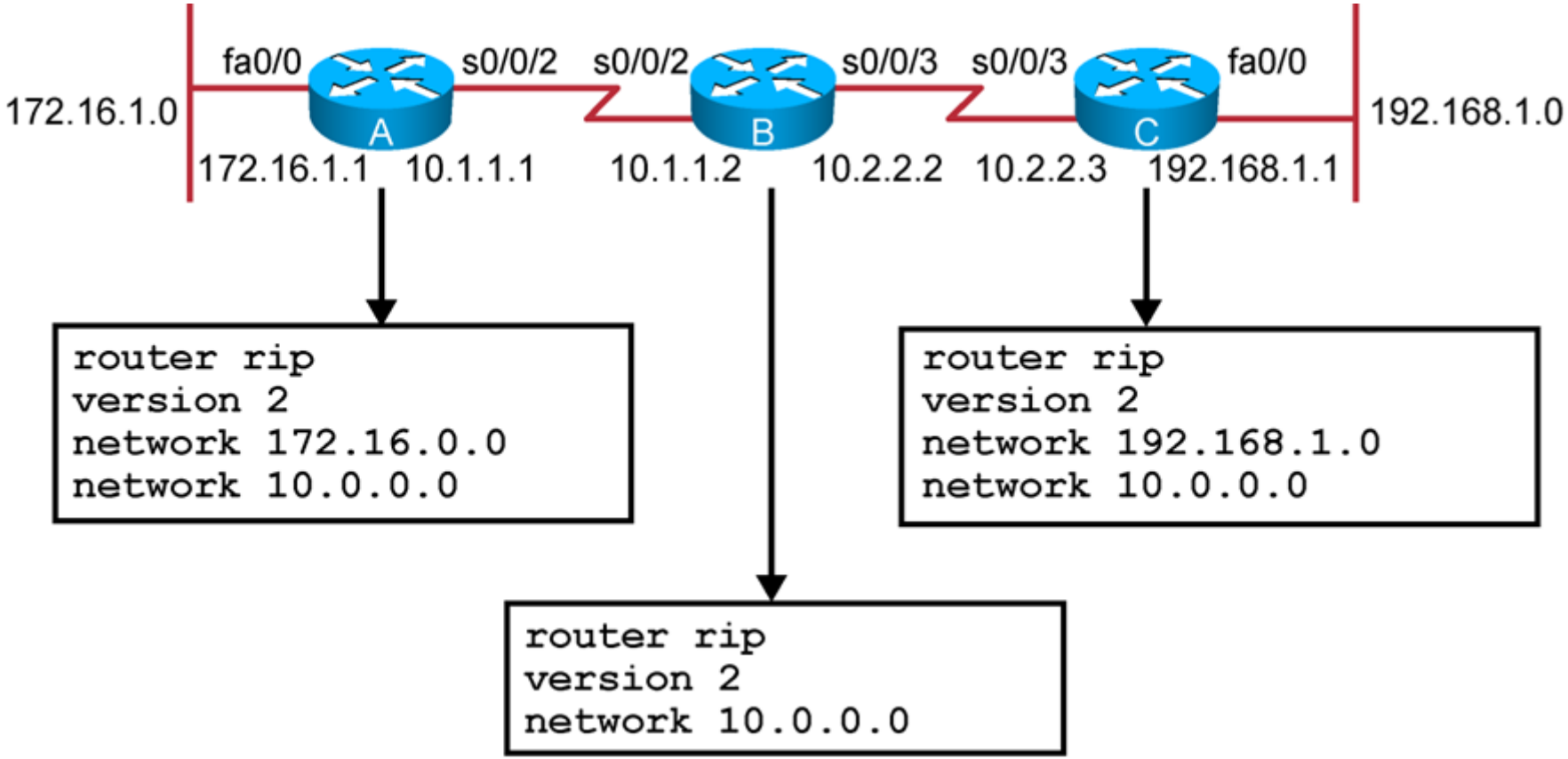
- Enables RIP version 2

```
RouterX(config-router)# network <major-network>
```

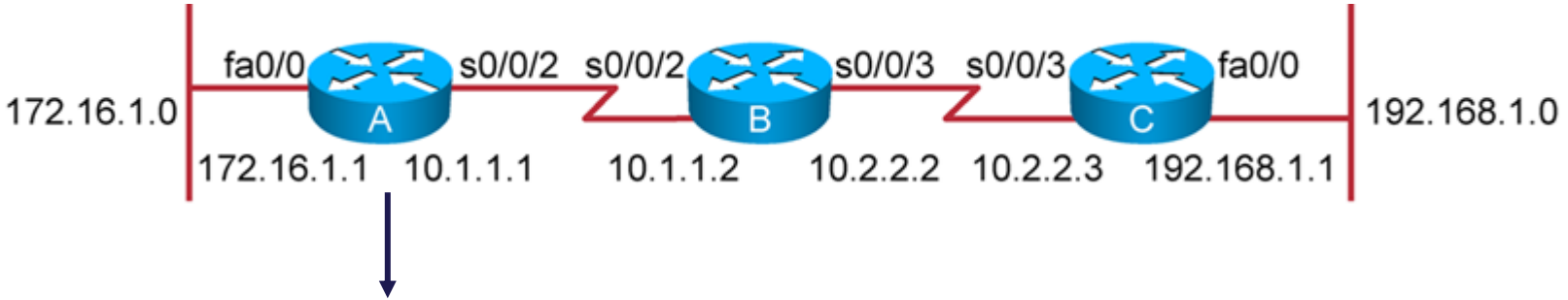
- Selects participating attached networks
- Requires a major classful network number



RIP Configuration Example



Verifying the RIP Configuration

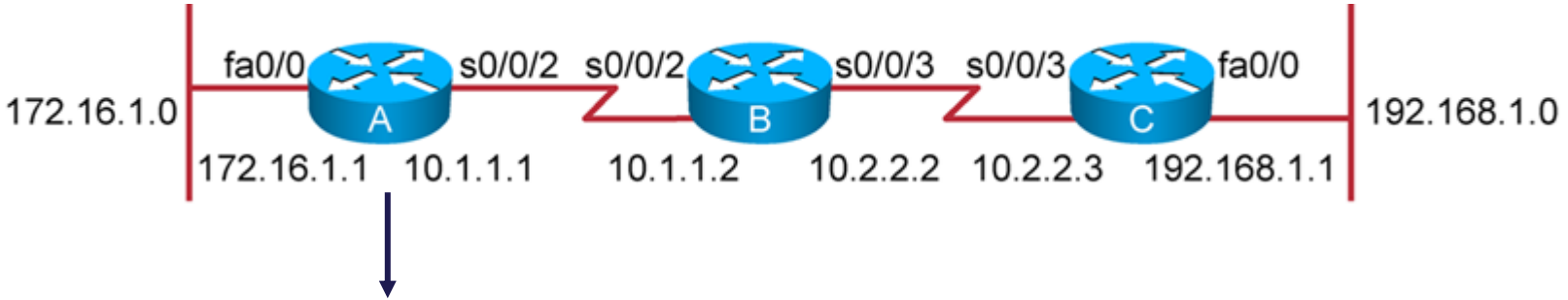



```
Routing Protocol is "rip"
  Sending updates every 30 seconds, next due in 6 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Redistributing: rip
  Default version control: send version 2, receive version 2
    Interface             Send  Recv  Triggered RIP  Key-chain
    FastEthernet0/0        2     2
    Serial0/0/2            2     2
  Automatic network summarization is in effect
  Maximum path: 4
  Routing for Networks:
    10.0.0.0
    172.16.0.0
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.1.1.2              120      00:00:25
  Distance: (default is 120)

RouterA#
```




Displaying the IP Routing Table

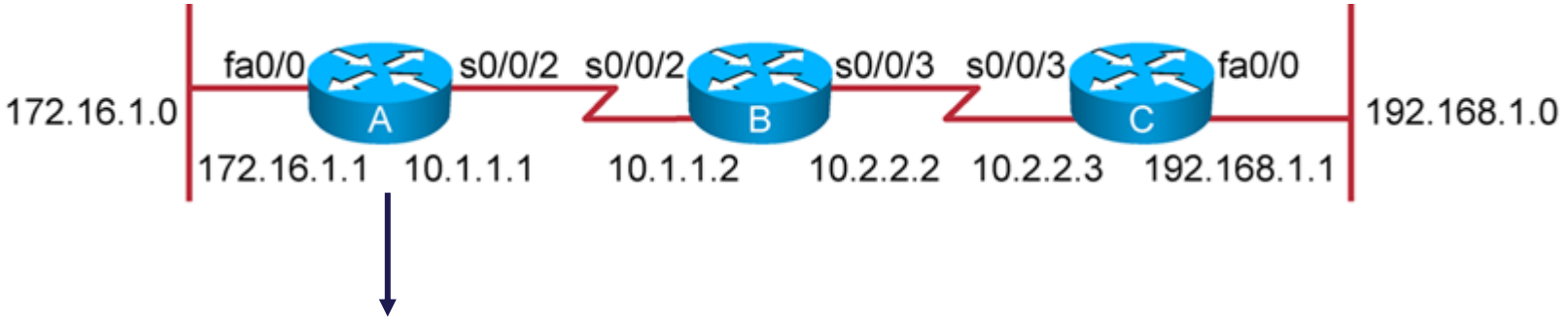



```
RouterA# show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, * - candidate default
       U - per-user static route, o - ODR
       T - traffic engineered route

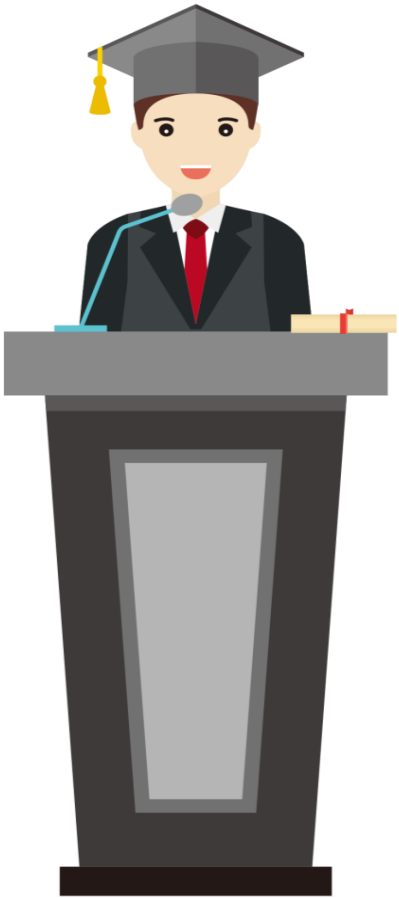
Gateway of last resort is not set


     172.16.0.0/24 is subnetted, 1 subnets
C       172.16.1.0 is directly connected, fastethernet0/0
     10.0.0.0/24 is subnetted, 2 subnets
R       10.2.2.0 [120/1] via 10.1.1.2, 00:00:07, Serial0/0/2
C       10.1.1.0 is directly connected, Serial0/0/2
R    192.168.1.0/24 [120/2] via 10.1.1.2, 00:00:07, Serial0/0/2
```

debug ip rip Command




```
RouterA# debug ip rip
RIP protocol debugging is on
RouterA#
00:06:24: RIP: received v1 update from 10.1.1.2 on Serial0/0/2
00:06:24:      10.2.2.0 in 1 hops
00:06:24:      192.168.1.0 in 2 hops
00:06:33: RIP: sending v1 update to 255.255.255.255 via FastEthernet0/0 (172.16.1.1)
00:06:34:      network 10.0.0.0, metric 1
00:06:34:      network 192.168.1.0, metric 3
00:06:34: RIP: sending v1 update to 255.255.255.255 via Serial0/0/2 (10.1.1.1)
00:06:34:      network 172.16.0.0, metric 1
```



THANK YOU!

